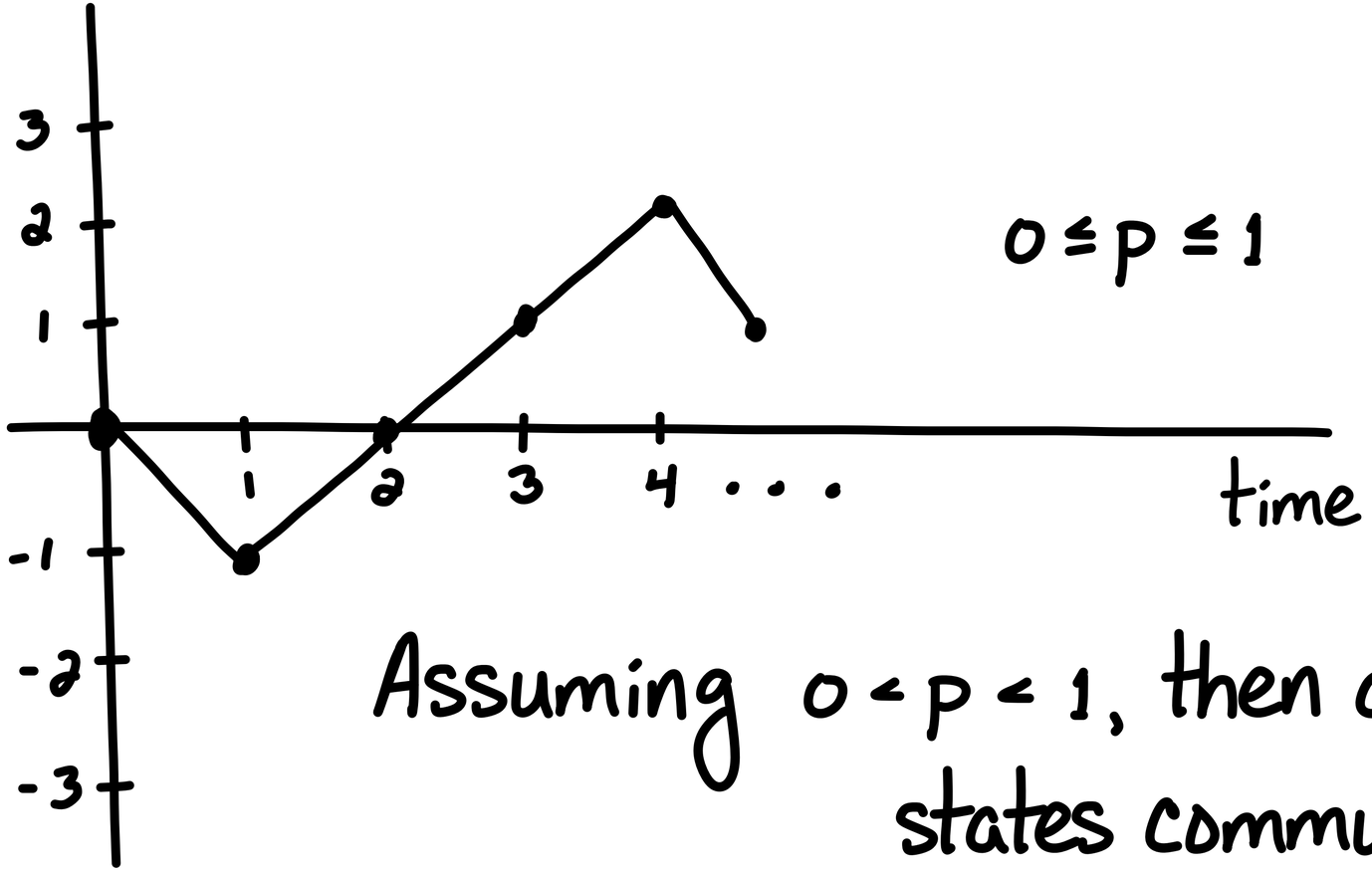
A "Random Walk" on the Integers

- Start with an initial value, say $X_0 = 0$.
- At each time step, either
  - go up one with probability p
  - go down one with probability $1-p$



$0 \le p \le 1$

Assuming $0 < p < 1$, then all states communicate!

$\Rightarrow$ all states are either recurrent or transient

- Check state 0.

i.e. Check whether $\sum_{n=1}^{\infty} P_{00}^{(n)} = \infty$ or $< \infty$.

- Note that $P_{00}^{(1)} = 0$.

More generally, $P_{00}^{(n)} = 0$ for $n = 1, 3, 5, \ldots$

- $P_{00}^{(2)} = p \cdot (1-p) + (1-p) \cdot p = 2p(1-p)$

$$P_{00}^{(2n)} = \binom{2n}{n} p^n \cdot (1-p)^n$$

$$\sum_{n=1}^{\infty} P_{00}^{(n)} = \sum_{n=1}^{\infty} P_{00}^{(2n)} = \sum_{n=1}^{\infty} \binom{2n}{n} p^n \cdot (1-p)^n$$

- Stirling's Approximation:

$$n! \sim \sqrt{2\pi}\, n^{n+1/2} e^{-n}$$

- Use the Limit Comparison Test from Calculus!

Limit Comparison Test

Consider series $\sum a_n$ and $\sum b_n$ with $a_n \geq 0$ and $b_n > 0$ for all n.

Define $c = \lim_{n \to \infty} \frac{a_n}{b_n}$

If $0 < c < \infty$ then both series either converge or diverge.

$$P_{00}^{(2n)} = \binom{2n}{n} p^n \cdot (1-p)^n$$

$$\sum_{n=1}^{\infty} P_{00}^{(n)} = \sum_{n=1}^{\infty} P_{00}^{(2n)} = \sum_{n=1}^{\infty} \binom{2n}{n} p^n \cdot (1-p)^n$$

- Stirling's Approximation:

$$n! \sim \sqrt{2\pi}\, n^{n+1/2} e^{-n}$$

- Use the Limit Comparison Test from Calculus!

- Will get

$$\sum_{n=1}^{\infty} P_{00}^{(n)} = \infty \quad \text{if} \quad p = 1/2 \quad \text{and}$$

$$\sum_{n=1}^{\infty} P_{00}^{(n)} < \infty \quad \text{if} \quad p \neq 1/2.$$

i.e. State 0 is recurrent if $p = 1/2$ and transient if $p \neq 1/2$.

⊛ The entire random walk chain is recurrent if $p = 1/2$ and transient if $p \neq 1/2$.

Suppose that $p = 1/2$. (recurrent)

- Starting at state 0 (for example), you might get a lot of "up moves".
- Suppose you reach state 10 (for example). The chain is going to want to want to "hover" there.

It may be difficult to return to 0.

We know that the chain will eventually return to 0 but it may take a long time!

Actually, the expected number of time steps to return is infinite!

???

- Let $T_0 = \min\{n \geq 1 : X_n = 0\}$.

Note that $P(T_0 < \infty \mid X_0 = 0)$

$$= \sum_{n=1}^{\infty} P(T_0 = n \mid X_0 = 0)$$

- However, $E[T_0 \mid X_0 = 0]$

$$= \sum_{n=1}^{\infty} n \cdot P(T_0 = n \mid X_0 = 0)$$

(This could blow up!)

<u>Definition</u>: Let i be a recurrent state.

- If, starting at i, the expected number of steps to return to i is finite, i is said to be "<u>positive recurrent</u>".

- Otherwise, i is "<u>null recurrent</u>".

We can show that positive recurrence (and therefore null recurrence) is a class property.

For an irreducible chain, we will call the entire chain transient or recurrent and recurrent chains will be either positive recurrent or null recurrent.

The Random Walk

- $p \neq 1/2 \Rightarrow$ the chain is transient
- $p = 1/2 \Rightarrow$ the chain is <u>null</u> recurrent